

Speech technology for telecommunications

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Speech is the easiest, most expressive and most natural means of human communication. Most of us have received intensive training in using it from the day we were born!. But speech is more than just a way of transmitting words or ideas — it conveys the essence of human emotion, moods, and personality. It is BT's core business, accounting for over 90% of revenues. It is also our primary means to access both the 26 million customers of the UK telephone networks, and around a half a billion telephone users world-wide. This paper introduces the key speech technologies, which are described in detail in the associated papers of this issue, and makes some personal predictions about future trends and challenges in this important, exciting and far-reaching field.

1. Introduction

Humans have been talking to each other using some form of speech for the last 50 000 years so it is not surprising that most of us take it for granted. Yet speech is an extremely complex process, and the simplest sentence contains a world of information besides its literal content. A statement can convey such nuances as the mood, approximate age, education, background and gender of the speaker. Not surprisingly, getting a machine to generate such subtle speech is somewhat of a challenge; making machines which can understand it is even more daunting, it is this challenge - to understand the mechanisms of speech production and perception — that forms the focus of this special issue of the BT Technology Journal.

The papers in this issue form a number of logical groupings — coding [1, 2], synthesis [3—5], recognition [6, 7] and performance assessment [8, 9]. The papers on implementation [10], spoken language processing [11], and on large-vocabulary interactive speech systems [12] provide an insight into the challenges of integrating speech technologies to provide network solutions that meet customer requirements. The paper by Attwater and Whittaker [12], in particular, provides a useful link to the following special issue on speech applications [13]. Another paper addresses important aspects of human perception of speech and other signals [14], and the paper by Lewis and Westall [15] boldly takes a leap forward to speculate on the future of telephony.

In each group of papers, tutorial material is given to help the reader, a perspective on applications is provided, and a

view from the 'leading-edge' of each technology area is presented.

This paper introduces the key technologies that underpin the field, and sets the scene. It provides a snapshot of current interactive speech applications, systems and issues, and concludes with some personal predictions on future trends and challenges.

2. Evolution

Although digital techniques have been pre-eminent in recent years, speech processing itself is not a new subject. Early voice coders, such as Dudley's VODER, date back to the late 1930s, and Von Kempelen made a voice synthesiser as early as 1791. The mathematics on which many current speech processing systems are based was established over 20 years ago, with the work of Atal, Makhoul and Itakura-Saito on linear prediction in the 1960s and early 1970s.

However, it was the advent of the digital signal processing microcomputer in the late 1970s that helped to convert research into practical, cost-effective systems, and heralded an upsurge of interest in new telecommunications applications [16, 17].

Figure 1 illustrates the evolution of digital signal processing (DSP) devices, measured in millions of operations per second (note the logarithmic scale). Also shown are the processing requirements for some 'popular' speech algorithms. Apart from the large-vocabulary speech recognition systems, most speech (and an increasing number of video) algorithms could be implemented on a

single device by 1995. It is interesting to note that, if aeronautical engineering had progressed at the same pace as device technology, it would now take under a minute to fly across the Atlantic!

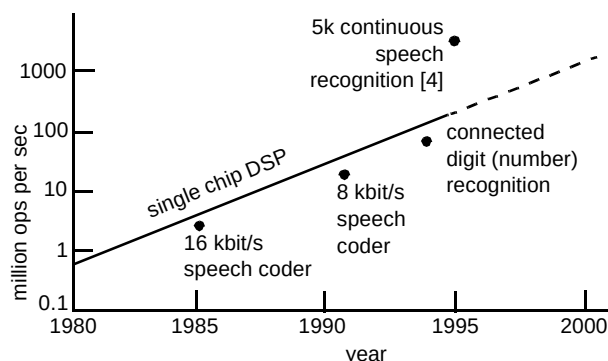


Fig 1 Evolution of DSP technology.

Telephony applications are today in the vanguard of this minor revolution, just as, to a lesser extent, modems were a few years ago. The latest digital mobile, aeronautical and multimedia terminals all depend on speech coding. The BT intelligent network answering service, CallMinder™, uses speech recognition to allow customers without a touch-tone phone to access the service, and speech coding to store efficiently the vast amounts of data generated. Speech echo cancellers are regularly used on international calls to facilitate two-way, simultaneous, speech conversations. Synthesised speech is used for services as diverse as the speaking clock, operator services, and reading e-mail over the phone.

Speech technology is an amalgam of many disciplines, as shown in Fig 2, and real-world speech applications increasingly require a broad-based, holistic approach to realise systems that are acceptable to the public at large. Engineering and computing need to be complemented by expertise in man-machine interaction, human perception,

psychology, acoustics, linguistics, natural language processing, and many other disciplines.

3. Applications and opportunities

Speech technology provides significant revenue opportunities by introducing new speech processing services which can easily be accessed over the telephone, twenty four hours per day, seven days per week [18, 19]; speech technologies can add value by:

- stimulating additional call revenues (e.g. the BT network-based call answering service, CallMinder™),
- creating new speech services (e.g. aeronautical telephony, talking e-mails),
- providing differentiation in existing services (e.g. speech enhancement, noise control),
- reducing operating costs (e.g. by partial or full automation of operator services and call centres),
- allowing control of systems by voice to make them easier to use and hence improving customer satisfaction,
- extending system usage to areas where other digit entry systems fail, where hands-free input is necessary, or where users require easily remembered access codes (e.g. names),
- providing a simple means of getting large amounts of variable text data to users over the telephone (e.g. text-to-speech).

Current activities are primarily directed towards applying the technology to the telephone network for interactive speech services such as:

- Voice store and forward
— CallMinder

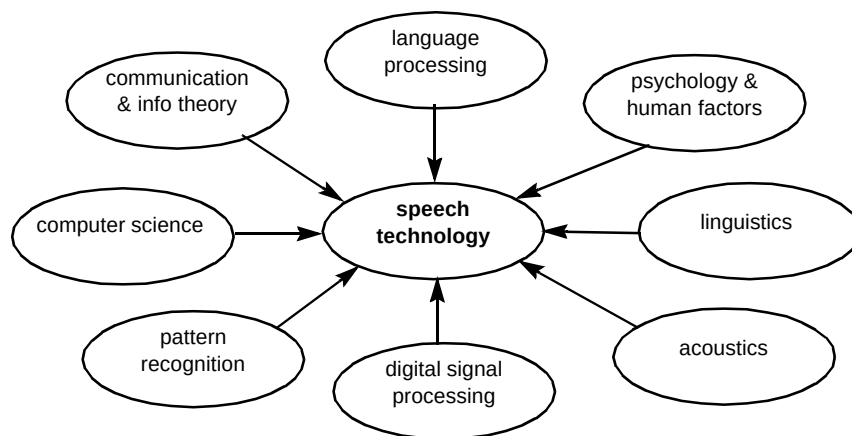


Fig 2 Speech technology disciplines.

- Finance
 - banking
 - stocks and shares— insurance quotations
 - credit card transactions
- Entertainment
 - betting
 - horoscopes
 - games
- Information services
 - timetables
 - Yellow Pages
 - News
- Telemarketing
 - promotions
- Teleshopping/reservations
 - theatres
 - airlines
 - catalogue shopping
- Field operations
 - data operation and retrieval
 - field personnel job despatch
 - voice access to electronic mail
- Automatic operator
 - network services
 - call centres

Since the core speech technologies are largely based on current understanding of human speech production and perception, and share some of its terminology, it is worth briefly reviewing how people generate and hear speech.

4. Principles of speech production and perception

Spoken language is communicated through the air via a longitudinal pressure wave. It is an extraordinarily effective method of transferring information between humans and has the capacity to reliably communicate complex ideas and emotions under constantly varying and hostile environmental conditions. It can also be viewed as an efficient method of 'encoding' linguistic information for verbal communication.

4.1 Speech production

The source for most speech sounds is air expelled from the lungs through muscular action (see Fig 3).

Air leaving the lungs passes through a body of interlocking cartilage called the larynx. Within the thyroid and cricoid cartilage that form the larynx are two bodies of muscle and membranous tissue called the vocal folds. During normal breathing the vocal folds are held apart allowing air to pass freely through the gap between the two folds (the glottis).

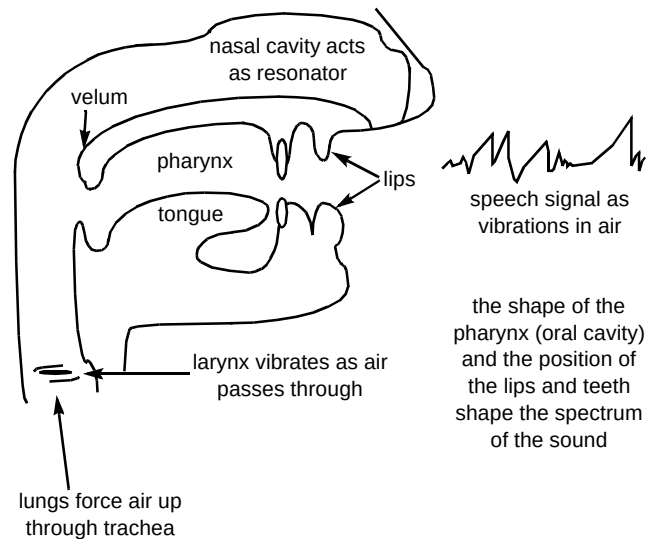


Fig 3 Speech production organs.

During 'voiced' speech the vocal folds are repeatedly brought together (a process called phonation). The range of pitch for adults is typically two octaves with the range for females being about an octave higher than for males. In contrast to voiced sounds, 'voiceless' or 'unvoiced' sounds are produced when the vocal folds are sufficiently abducted to allow air to pass relatively unimpeded through the glottis, as when producing sounds such as 's' and 'ch'.

The acoustic signal, generated by the larynx, is then modified by the vocal tract by manipulation of the position of the velum, tongue, teeth, jaw and lips to determine which sound will be radiated. 'Plosives' are formed by closing off the vocal tract by the tongue, lips or teeth and then releasing the sound energy ('p', 'b' and 't' are examples). Although the vocal tract is capable of producing an infinite number of distinct sounds, at the linguistic level the basic perceptual unit of speech is the phoneme which depends on the word being spoken and the position of the phoneme within the word. The English language consists of 44 phonemes, as indicated in Table 1.

This 'source-filter' speech production model has an electrical analogy, as shown in Fig 4. This system was first implemented by Homer Dudley of Bell Labs for the World Fair of 1939. Trained operators controlled the various parameters of the filter and excitation to produce human like sounds. This model formed the basis for most synthesis and vocoder systems for the next half a century, and remains valid (section 6.1).

4.2 Speech perception

Unlike the vocal tract, which in addition to sound production is also used for breathing, tasting and eating, the ear seems to be custom-built for the purpose of detecting and

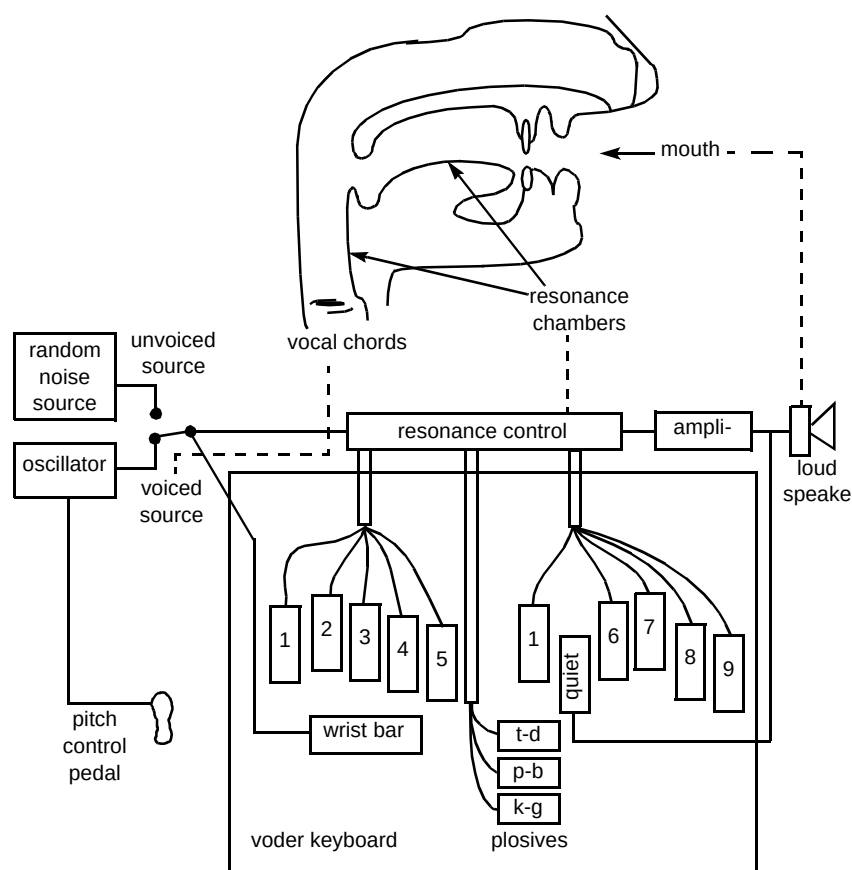


Fig 4 Dudley's VODER (1939).

analysing sounds. The auditory system consists of three sections -- the outer, middle and inner ears, as shown in Fig 5.

Table 1 SAM-PA (speech assessment methodology phonetic alphabet) phonetic symbols for transcribing english.

p	<u>p</u> et	t	<u>t</u> ie	k	<u>k</u> ey
b	<u>b</u> ag	d	<u>d</u> og	g	<u>g</u> ap
m	<u>m</u> ap	n	<u>n</u> ap	N	<u>r</u> ing
f	<u>f</u> ive	v	<u>v</u> ia	T	<u>th</u> igh
D	<u>th</u> ey	s	<u>s</u> ixes	z	<u>z</u> oo
S	<u>sh</u> ake	Z	<u>l</u> ounge	l	<u>l</u> ip
w	<u>w</u> ill	r	<u>r</u> ing	j	<u>y</u> ou
h	<u>h</u> it	tS	<u>ch</u> urch	dZ	<u>j</u> ive
i	<u>b</u> ead	I	<u>b</u> it	e	<u>b</u> ed
(	<u>b</u> ad	A	<u>h</u> ard	Q	<u>c</u> od
O	<u>c</u> ourt	U	<u>g</u> ood	u	<u>f</u> ood
V	<u>b</u> ud	3	<u>b</u> ird	@	<u>a</u> go
eI	<u>d</u> ay	@U	<u>m</u> ow	aI	<u>h</u> igh
OI	<u>t</u> oy	I@	<u>h</u> ear	e@	<u>r</u> are
u@	<u>R</u> uhr	aU	<u>c</u> ow		

The outer ear has two roles — it aids sound localisation by altering the spectrum of the sound, depending on the direction of the source, and it transmits sounds to the

tympanic membrane. The tympanic membrane, or eardrum, vibrates under the influence of sound pressure waves, much like a diaphragm in a microphone. The middle ear consists of three small bones, known as the ossicles which transfer vibrational energy to the inner ear.

The inner ear is a system of fluid-filled cavities and the oscillations of the stapes set up a pressure wave, which causes the fluid in the cochlea to move.

The cochlea not only converts mechanical vibrations into nerve signals but also selects the frequency of the incoming sounds — a direct physiological example of a set of filters. Contained in the cochlea is the basilar membrane which is graded in width and stiffness along its length, being narrow and stiff at the basal end, and wider and more flexible towards the apex. The auditory transducer which turns sound energy into nerve stimuli is the 'organ of Corti', which is attached to the basilar membrane. It contains a large number of specialised and complex hair cells which perform the transduction process between the mechanical vibrations of the basilar membrane and nerve impulses transmitted to the brain.

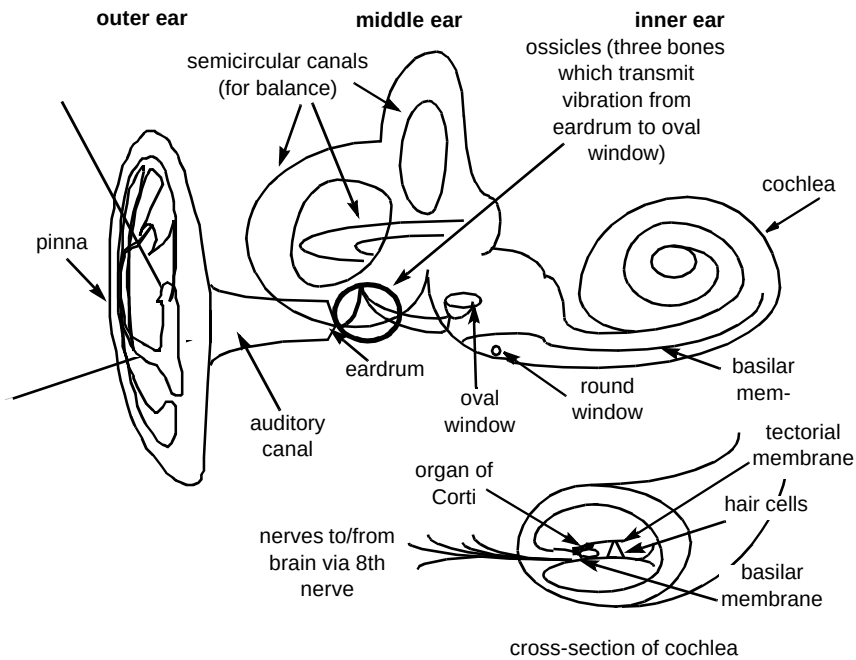


Fig 5 Physiology of the ear.

4.3 Speech signals

In Fig 6a a typical speech waveform is shown as a function of time, and Fig 6b shows the corresponding frequency response. Frequency domain representations have been used for a number of years, which is not surprising given that the vocal tract acts as a resonating cavity with resonant characteristics that vary relatively slowly as the articulators move from one sound to another.

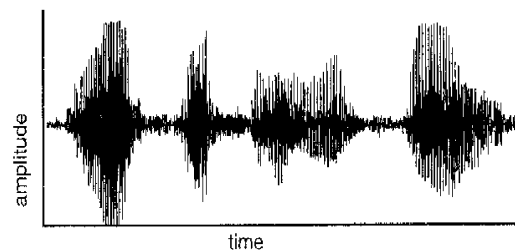


Fig 6a Speech signal.

A device used to display this is known as a spectrograph, as shown in Fig 6b. Here the vertical axis shows frequency, the horizontal axis is time and the darkness corresponds to sound intensity. Clearly visible in this plot are the peaks representing the dominant resonances of the vocal tract, known as formants.

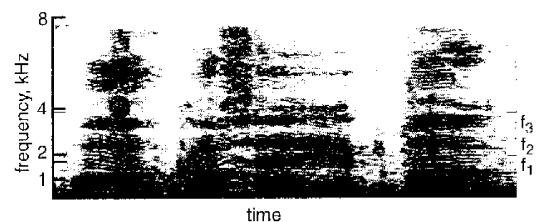


Fig 6b Wideband spectrogram.

4.4 The mutability of speech

For many years speech was considered to be composed of linear sequences of elemental speech particles, or phonemes, put together rather like a string of beads. This misleading interpretation of the speech signal led engineers to imagine that machines could recognise speech simply by decoding the individual phonemes as they appear through time, in the same way that a modulated data signal could be produced.

In a similar vein, it appeared that synthetic speech could be produced by simply storing an example of each of the phonemes of a language and then sticking them together to form the new word. Unfortunately this proved not to be the case. While it is convenient to visualise speech as a set of discrete symbols, in reality linguistic information is not encoded in discrete packets of time as imagined by this model, and traditional methods of dealing with coded

signals will not work with speech. Our familiarity with text also encourages us to think of speech as 'atomic'; but, convenient as this picture is, speech is not like text — see Fig 7.

There are no gaps between words.

Sometimes there are gaps in the

middle of words. Accents,

dialects and Stress *eggs*ist

Fig 7 Example of speech as text

Compared with many other areas of digital signal processing, the challenges presented by speech processing are immense. Although the public imagination has been fired with images like 'HAL' in Stanley Kubrick's '2001 — a Space Odyssey', the goal of producing complete natural language interfaces between humans and machines is still a long way off. The difficulty arises due to the variability in factors that dictate performance:

- the great variety in the signal characteristics, when the word or phrase is uttered by different speakers and even when repeated by the same speaker,
- the wide variety of channel characteristics through which the speech signals are sent,
- the nature of the accompanying background noise.

Above the level of the acoustic signal, ambiguities can occur at a higher level of linguistic abstraction. The phrase 'It's easy to recognise speech' could be interpreted as 'It's easy to wreck a nice beach' at a conference of surfers! To make matters worse, people do not always say what they mean or mean what they say.

This mutable behaviour is unavoidable and must be tolerated by a speech processing system which aims to provide a seamless and natural interface between people and machines.

Before introducing the core speech technologies, it is worth reviewing, at a system level, how these components are combined to create an application. This is the subject of the next section.

5. Interactive voice response systems

Developments in speech technology have enabled a new generation of interactive voice response (IVR) services operating over the telephone network. As indicated in section 3, these range from telco-type services such as automation of directory enquiries, to customer-handling and information retrieval applications which can offer commercial opportunities for many companies. A major activity is the integration of speech technology with existing databases, IT processes and call-centre capabilities. These developments range in scale from small bespoke systems involving single-line PC solutions to embedded network systems capable of supporting many hundreds or thousands of simultaneous calls. The larger scale applications can significantly re-engineer the way in which customers interact with providers of information, goods and services.

A typical interactive speech system is shown in Fig 8.

Speech input is achieved via the recogniser (which may be supported by DTMF or dial-pulse input), and speech output is by means of the synthesiser (which can be based on stored concatenated speech or text-to-speech). For a network-based application serving thousands of customers, a speech coder may be necessary in order to make efficient use of expensive storage media. The user aims to extract information from the system database (or voice store) by means of spoken commands.

Until recently, the performance of commercial speech recognisers restricted their use to relatively small vocabularies unsuited to applications where information such as names and addresses is required. However, the recent emergence of accurate large vocabulary recognition at acceptable cost has increased the range of practical applications.

The information manager interfaces with the host application or database and acts as a mediator for high-level dialogue requests, for example, building recognition vocabularies based upon the database content. The role of the information manager is discussed by Attwater and Whittaker [12].

The dialogue controller ensures that the required information is obtained from the user in a controlled and structured way. It acts as a kind of gearbox matching the recogniser performance (engine) to the application (the wheels). Like any gearbox it can, to some extent, overcome deficiencies in the engine, but only if the engine is basically sound. Dialogues are not a substitute for inadequate or inappropriate speech technology.

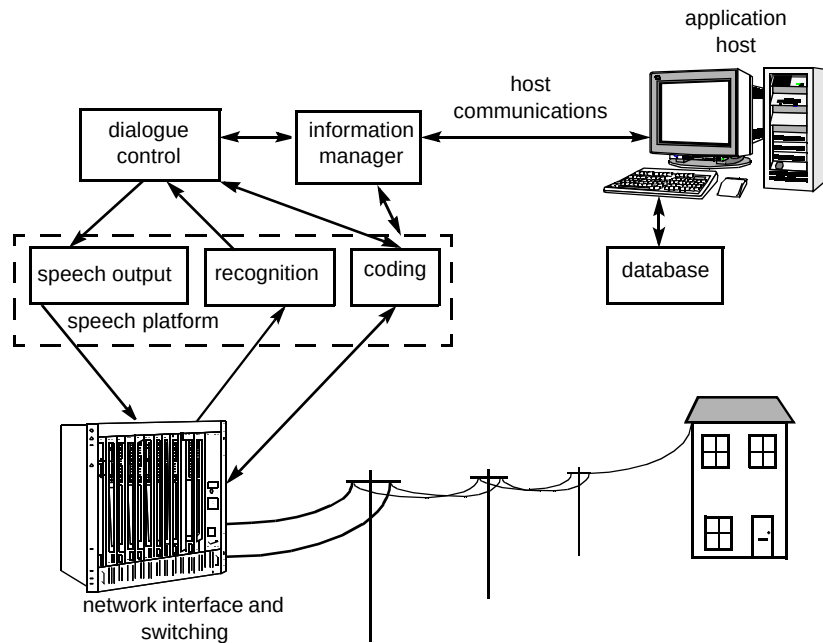


Fig 8 Interactive speech system.

5.1 Dialogue design

Dialogue design is often the key to successful IVR speech system design [12]. It defines the system interface which the caller will encounter, guiding and interpreting each user interaction and selecting the appropriate response. The dialogue controller also ensures that the system recovers gracefully from any errors which are detected in the speech recognition, either by asking the user to repeat, or by modifying the dialogue to obtain the required information by rephrasing the question.

From the user's point of view the system should be friendly, reliable and comfortable to use. To achieve this, it is necessary to ensure each component of the system performs adequately, and aligns with the customer's expectations. It is a considerable challenge to orchestrate all the constituent technologies to best meet the expectations of the whole user population under all operating conditions.

5.2 Example of a dialogue

Within the UK, Directory Assistance (DA) is a chargeable service with a database of over 20 million residential, business and government entries involving nearly half a million distinct surnames, and nearly 30 thousand distinct localities. An example of the dialogue for such an experimental application is given in Attwater and Whittaker [12]. Partial spelling of the first N letters helps the recogniser distinguish between confusable names and homonyms (words which sound the same but which are spelt differently — typically 30% of names may be homonyms or synonyms of others).

In current IVR systems, the flow of the dialogue is controlled primarily by the system, not the user. Information is garnered one item at a time and a rigid dialogue structure enforces certain strategies for confirmation, rejection, error detection, correction and reprompting.

It can be frustrating for users to find themselves 'steered' through a complex dialogue with a machine only to find either that they are lost or that they have ended up with the wrong piece of information. This calls for a different approach to dialogue design, and systems are beginning to emerge where it is possible for the caller to respond much more spontaneously, flexibly and take control of the interaction. For example, the next generation of interactive voice response system might have the following 'conversation' with you as shown in Fig 9.

How many messages do I have from Denis?
There are two messages from Denis Smith and one from Denis Crowe
Can I hear the ones from Smith?
The first message from Denis Smith is about 'Budgets', it's 20 seconds long and a summary is ' ...
(Interruption) ...'
Give me the next one....
The next message is about 'The future of speech recognition', it's 10 seconds long and a summary is ..
OK, — uhm — read the previous one back to me in full.

Fig 9 Example of possible future interactive voice system 'conversation'.

The example in Fig 9 illustrates a number of features which are desirable in future interactive voice applications.

The system does not need to constantly prompt the caller for explicit information. It uses discourse and domain knowledge to resolve ambiguous expressions such as 'give me the next one' and is able to understand references to earlier information. The caller can interrupt (bargue in) at any stage, and the interrupting phrase is recognised. The caller controls whether the full message is heard (having been told beforehand how long it is), or if a summary is given. The system discards paraverbals such as 'uhms' but is able to identify words of significance wherever they occur (wordspotting).

Such a dialogue allows the initiative in the conversation to migrate from the computer to the user. More fluent interactions of this type will inevitably place greater demands on the underlying speech technology and require more language knowledge than would be the case with more structured dialogues. However, spoken language will be an essential vehicle for searching and retrieving information, and may help to make it accessible to all users, of all ages, and not just the computer literate.

For example, cellphones are already too small to support a keyboard. For such applications, interactive dialogue-based systems will require both high-performance speech recognition and language understanding in order to properly identify and respond to user's requests. These, and other aspects of spoken language processing are discussed by Wyard [11].

5.3 The speech applications platform

To address network-based applications for interactive voice systems, BT has developed the speech applications platform (SAP). This is a modular system based on industry standards for hardware, software, interfaces and protocols. The SAP can be scaled from single line systems to large embedded network and intelligent network installations and supports a range of exchange signalling and network integration options. Figure 10 shows the speech applications platform connected as a network element.

The SAP provides a wide variety of speech and signal processing facilities such as large-vocabulary speech recognition, text-to-speech, speech coding, tone and loop disconnect recognition. The system is closely integrated with BT's billing, maintenance, customer and network management systems.

Dynamic allocation of individual speech functions on call establishment ensures that efficient use is made of available DSP resources, and that these are made available to other channels when no longer in use. Applications may be written for the SAP using an object oriented programming interface or through an integrated graphical dialogue creation tool [13, 20].

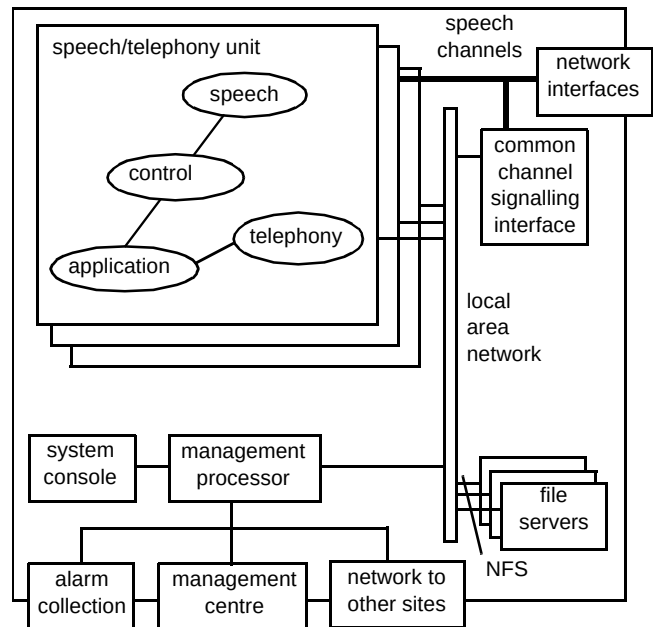


Fig 10 SAP as a network element.

6. Speech technology

The key underlying speech technologies for IVR systems are now reviewed, starting with the most mature of the technologies — speech coding.

6.1 Speech coding

This is the process of converting speech into digital bit streams for efficient storage and/or transmission over bandlimited channels. It can exploit redundancy in speech signals to reduce the transmitted bit rate while at the same time exploiting the known properties of human speech perception to reduce the coding distortion to acceptable levels. Speech coders for particular applications are selected according to a trade-off between coding complexity, bit rate and signal quality. The design trade-offs are further complicated by additional factors such as:

- codec delay (which makes echoes more audible),
- transparency (which effects the ability of the codec to pass non-speech signals),
- coder robustness to transmission errors and background noise,
- tandeming ability (or the ability to mix different coders in a transmission path without the accumulation of distortion to unacceptable levels).

Speech coding has been subject to considerable international standardisation activities in recent years. Standards now exist for 32 kbit/s (G.721), wideband (7 kHz) speech coding (G.722), 16 kbit/s (G.728), 8 kbit/s (G.729), variable-rate speech coding for circuit-

multiplication equipment (G.726), wideband packet networks (G.727).

Speech coders are in demand where there is a need to conserve radio spectrum. Examples include aeronautical telephony via satellite (Skyphone — 9.6-kbit/s coder), digital cellular (GSM — 13-kbit/s coder) and digital cordless systems (CT2/DECT — 32-kbit/s coder). Barrett [2] provides a comprehensive review of speech transmission requirements and techniques for digital mobile radio applications (using GSM as a case study).

Coders have also been applied in customers' private network applications, e.g. for speech-and-data multiplexers, in PSTN and ISDN videophone applications for the audio channel (e.g. VC8000 [21]), and in interactive network-based voice-messaging applications (e.g. CallMinder) where cost of storage is still a critical factor [20].

A comprehensive review of speech coding algorithms and applications is provided by Wong [1]. Coders are usually classified into the following three types:

- waveform coding,
- vocoding,
- hybrid coding.

The aim of waveform coders, as the name implies, is to reproduce the original waveform as accurately as possible. As these coders are not speech specific they can deal with non-speech signals, such as background noise, music and multiple speakers, without difficulty. However, the cost of this fidelity is a relatively high bit rate, e.g. PCM, adaptive

differential PCM (ADPCM) and sub-band coding [22] techniques.

In contrast, vocoders (voice+coders) make no attempt to reproduce the original waveform, but instead derive a set of parameters at the encoder, which are transmitted and used to control a speech production model at the receiver (Fig 4). Typically, linear prediction coding (LPC) is used to derive the parameters of a time-varying digital filter, which models the dominant resonances of the vocal tract (section 4.1). Speech quality, although intelligible, tends to be synthetic and variable between speakers. Hence vocoding is not used for telephone network applications.

Hybrid coders combine features from both waveform coders and vocoders to provide good quality, efficient speech coding. At rates between about 16 kbit/s and 4 kbit/s, good quality coding is achieved using 'analysis by synthesis' techniques, as shown in Fig 11.

The objective is to derive an excitation signal, like that produced by the glottis (section 4.1), such that the difference between the input and synthesised speech signals is minimised according to some suitable perceptual criterion. A good description of the operation of this type of technique can be found in Boyd [23]. This technique is the basis of the coding schemes in Skyphone (9.6 kbit/s), the ITU low-delay standard (16 kbit/s), and the recent ITU standard for toll-quality coding at 8 kbit/s.

As shown in Fig 12, 'toll' or telephony-quality coders currently operate down to bit rates of around 8 kbit/s, with the prospect in sight within a few years of rates around 4 kbit/s. A new class of coders, based on sinusoidal transform

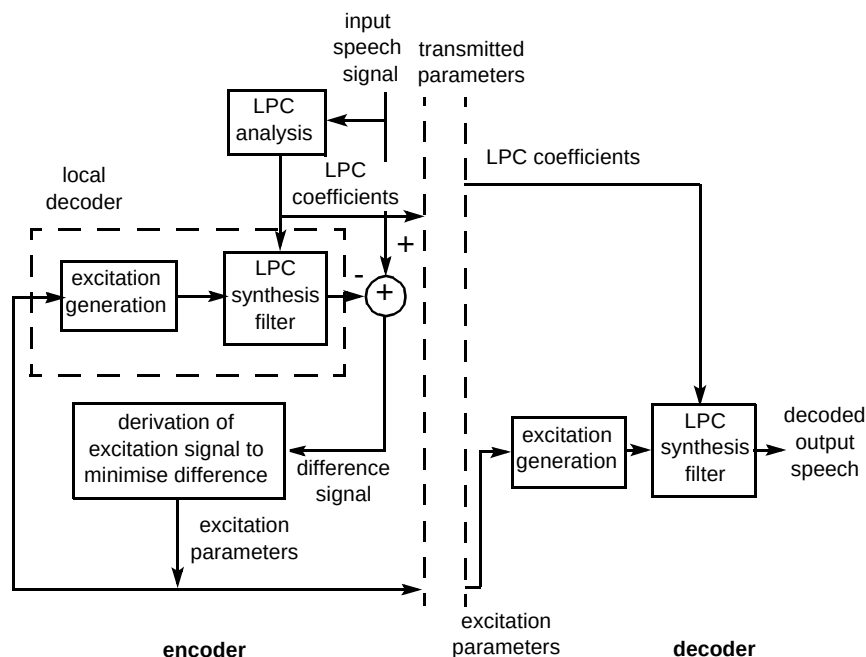


Fig 11 Analysis-by-synthesis coding scheme.

coding and waveform interpolation techniques appears to be especially promising at these lower rates. Wong [1] reviews these techniques.

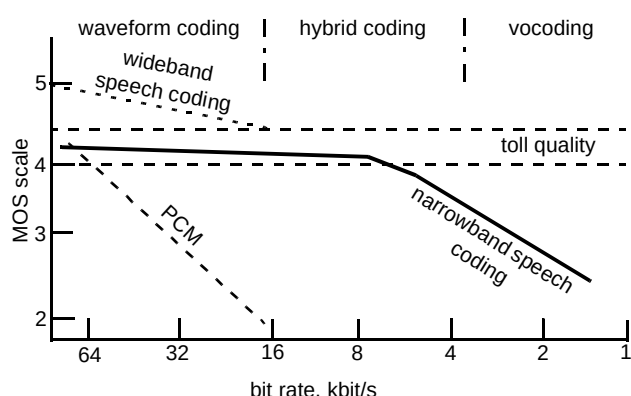


Fig 12 State-of-the-art in speech coding.

In the longer term, more significant gains are theoretically possible if one considers that the maximum rate of articulatory movement is limited by inertia of the physiological structures of tongue and jaw to about ten discrete sounds per second. Given the 44 phonemes of English (Table 1), this gives a theoretical minimum of around 100 bit/s (even allowing for non-verbal cues such as emotion and phrase emphasis to be included).

There is also considerable current interest in coding enhanced quality wideband speech (typically between 7kHz and 20 kHz bandwidth) at rates of around 1 bit/sample for speech and 2 bit/sample for audio (including music), for ISDN applications such as teleconferencing, for CD audio compression (MPEG-audio), and for commentary channel transmission.

6.2 Speech recognition

Today, interaction with computers normally involves the use of more artificial methods, such as keyboards, touchtone, touch-screens or mouse-driven menu systems. Speech recognition can improve the user interface by allowing spoken commands for accessing network services or for call routing, e.g. voice dialling over mobile phones.

Historical perspective

The earliest attempt at voice recognition was the voice-operated writing machine described by Flowers in 1916 [24]. The first paper on electronic speech recognition was published by Bell Labs in 1953 and described a system which could recognise single digit utterances spoken in isolation by a single speaker. Figure 13 shows this system, which used zero crossings from two frequency bands as features, and a form of vector quantisation of the track traced by the oscilloscope beam for the recognition decision. Later in the 1950s the technology was extended to

recognition of isolated syllables and phonemes in isolated words at RCA in the USA and University College, London. This pioneering work provided statistical information concerning possible phoneme strings in English and was the forerunner of stochastic language modelling used in modern large vocabulary systems. An early trial of speaker-independent recognition was held at MIT Lincoln Labs in 1959.

During the 1970s, dynamic programming (also known as dynamic time warping (DTW)) emerged to cope with the time expansion and contraction found in normal speech. Connected-word recognition was also developed, including the two-stage dynamic programming procedure, continuous speech recognition (CMU and IBM) and speaker-independent recognition (Bell Labs).

During the 1980s there was a shift from template (reference-pattern) matching to statistical modelling methods, such as hidden Markov modelling. Stochastic methods started to be used for language modelling in place of rule-based methods, and a recognition system based on word concatenation probabilities was investigated at IBM. Also in the 1980s the Defense Advanced Research Programme Agency (DARPA) drove forward continuous speech recognition with large vocabularies (1000+ words) at establishments such as SRI, MIT, Carnegie Mellon University, Bolt Beranek and Newman, and Lincoln Labs.

BT began work on speech recognition in the early 1980s resulting in Topaz — the first repertory dialler for hands-free voice dialling in cars. Early speaker verification and small vocabulary speaker-independent recognisers were also deployed in the pioneering trial of automated banking with the Royal Bank of Scotland in 1988 [24]. BT first demonstrated rapid vocabulary generation together with spelt connected recognition in 1993. More recently, large-vocabulary recognition has been implemented on the network-based speech application platform (see section 5.3) [20].

Currently telecommunications provides one of the largest market sectors for speech recognition, with applications as diverse as voice dialling, credit card validation, and access by voice to the information services indicated in section 3.

Recognition technology

A speech recogniser is designed to recognise one of a set of words or phrases specified in a vocabulary. This is a difficult task, and not even a person is 100% accurate.

Figure 14 shows the components of a speech recognition system. The front-end stage performs feature extraction, taking segments of the speech at regular intervals and transforming them to facilitate pattern

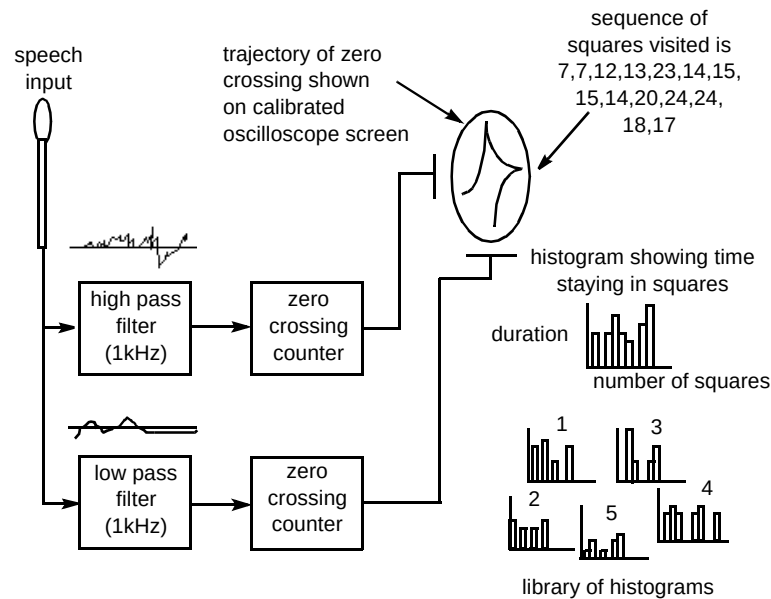


Fig 13 Bell recogniser (1953).

recognition. Part of the process involves nonlinear filtering operations, which are assumed to occur in the cochlea (section 4.2). The resulting frames of features are then processed, to identify either words or parts of words (phonemes). In a more complex scenario, knowledge of the context can be used to aid sentence recognition by exploiting semantic (meaning), syntactic (grammar) and dialogue (discourse) constraints.

Many current speech recognisers use the technique of hidden Markov modelling (HMM) for pattern matching. An HMM is a type of model based on a statistical representation, and this helps the recogniser to cope with most of the variability in the way people speak. The recogniser aims to identify, from observing a sequence of speech features, which of the stored Markov models is most likely to have produced these observations. A good 'layman' description of speech recognition techniques, including HMMs, is provided in Power [6]. For those who wish to delve deeper into HMMs, a good theoretical treatment of the subject can be found in Cox [25].

A connected-word recogniser has to match utterances to strings rather than to single words. This is usually achieved with a finite-state network of word models representing the whole vocabulary. The shortest path through this network which matches the utterance is selected as the recognised string.

Recognition performance depends on the vocabulary to be recognised and the number of users. A business dictation system has a large vocabulary (typically 20 000 to 30 000 words), but can be trained for a particular speaker and exploit tight constraints on language and word statistics.

In contrast, speech recognition for telephony is likely to use only a small vocabulary, since the recognition task is far more challenging — with much greater signal, network and speaker variability. Also context and grammar constraints tend to be weaker, as with surname and address recognition. As an example, BT's latest speaker-independent recogniser is capable of identifying several thousand words and any connected grammar over the telephone network. Scahill [10] addresses the implementation aspects of telephony-based speech recognition.

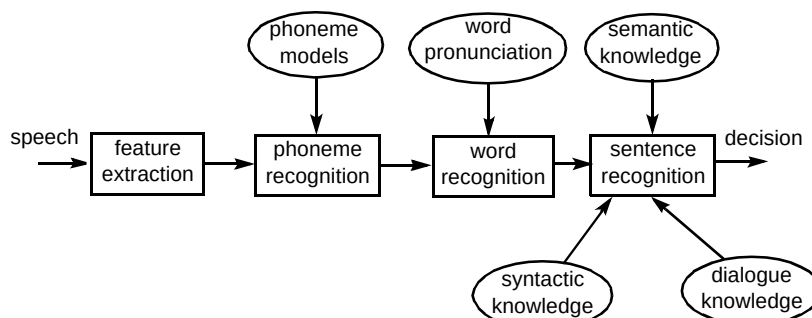


Fig 14 Modern speech recognition system.

In the past, the production of new vocabularies took several months, requiring many examples of the words from a hundred or more speakers. Also the whole process had to be repeated if a change or addition to the vocabulary was subsequently needed. Recently, a method has been developed to allow new vocabularies to be generated very quickly, by simply typing the words to be recognised. Word models may be built from phonetic transcriptions using either a dictionary or a set of letter-to-sound rules. Because of the various approximations, recognition accuracy for text-generated vocabularies are generally inferior to those which have been trained on actual speech utterances. In applications where the vocabulary is changing rapidly, such as directory or customer identification applications, the system must be able to dynamically update its active vocabularies. This aspect is discussed in detail by Attwater and Whittaker [12].

Speaker-dependent repertory dialling (e.g. 'Call Bob') has been available for some time. Typically, this requires the user to repeat the word or phrase a small number of times, under the guidance of the dialogue controller. Recently, this concept has been extended to speaker-independent operation, as described in Pawlewski [7].

6.3 Speaker recognition

Speaker recognition is used to identify or validate a speaker from a sample of his/her voice.

One of the first applications of speaker recognition was forensics. In 1660, speech was used as the key to detect a criminal for the first time in the case concerning the death of Charles I. However, at that time, the decision was made subjectively. In the 1940s, the sound spectrograph was invented by Potter at Bell Labs, and it became possible to draw voice prints (visible speech) as shown in Fig 6(b).

The first paper on speaker recognition was by Pruzansky of Bell Labs in the 1960s. From then until the early 1970s speaker recognition research was started at several laboratories, including IBM, TI and NTT, and both text-dependent and text-independent methods were investigated. In the 1980s, speaker recognition research also started to use stochastic approaches such as HMM.

The first network-based use of speaker recognition by BT was as part of a trial of automated banking with the Royal Bank of Scotland in 1988 (using dynamic time warping) [24].

Speaker verification technology can be used to validate the claimed identity of a person from his or her voice-print. It is often combined with a general-purpose connected speech recognition system to recognise a spoken PIN, and then to check that the PIN was spoken by the authorised person [7].

Speaker identification can be used to recognise a member of a closed user group (e.g. a family) directly from a known spoken word (text-dependent) or from any spoken utterance (text-independent).

Current applications include secure access control to information, banking, computer networks, PBXs and work areas. The technology can also be used to provide access to a range of network services according to customer profiles, such as name dialling and travel information.

6.4 Speech synthesis

The evolution of speech synthesis is shown in Fig 15.

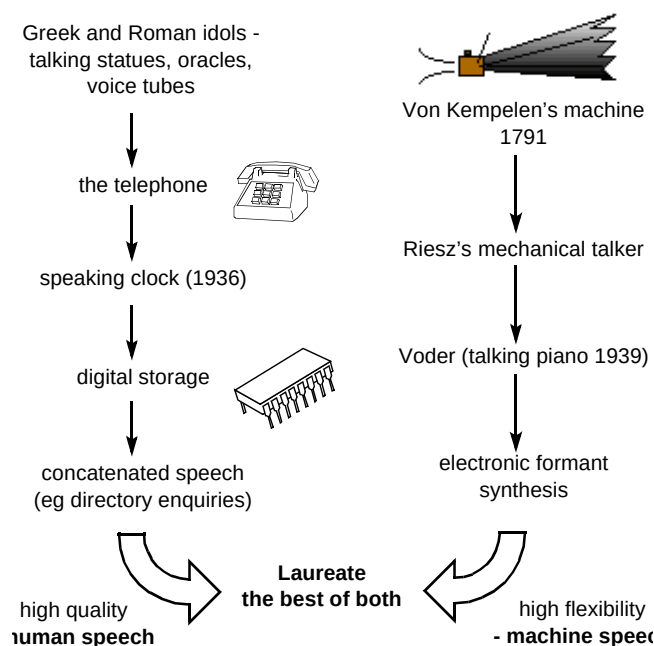


Fig 15 Evolution of speech synthesis

In 1791 Wolfgang Ritter Von Kempelen constructed a talking machine (Fig 16); it consisted of a bellows, a mouth shape, nostrils and whistles. The machine included a compressible leather tube and an air chamber equipped with a reed leading to a soft leather resonator which could be manually shaped for the formation of the vowel sounds. Consonants were created by holes which the 'player' closed by movement of the fingers. The Von Kempelen machine could produce about twenty different sounds.

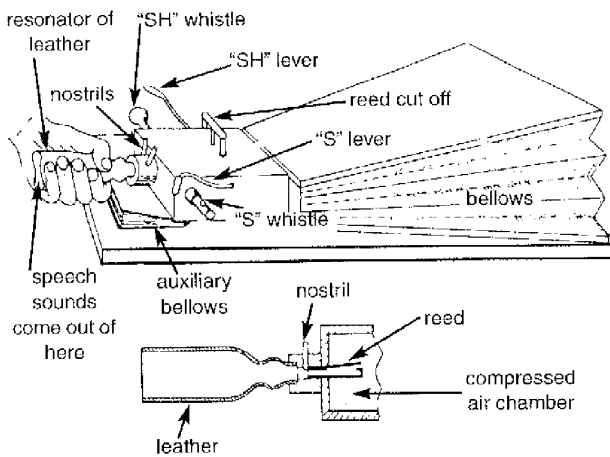


Fig 16 Wheatstone's Replica of Von Kempelen's machine

The earliest electronic synthesis of speech was achieved by Dudley in 1939 when he demonstrated a manually controlled speech synthesiser known as VODER (Voice Operated Demonstrator) at New York's World Fair (Fig 4). After the second world war, development began in the British Post Office of techniques to analyse and synthesise natural speech. This led to a variety of synthesis systems based on the source-filter model of speech production of which the formant synthesisers developed by the Joint Speech Research Unit [26] in the UK and at MIT in the USA are perhaps the best examples.

Speech output from computer-based equipment has been commonly achieved by generating messages from stored speech fragments (as with the BT speaking clock). These messages require the recording by a speaker and although a natural sounding voice output is achieved, significant amounts of data storage are required. An additional constraint on system design and extension arises when the messages need to be changed or updated if the original speaker is not available. There are many applications where the versatility of a full text-to-speech system is the only practical solution.

BT's synthesis system, known as Laureate, is designed to convert unrestricted textual input into speech. Laureate differs from conventional speech synthesis systems in that it does not attempt to artificially generate sounds to mimic human utterances (section 4.1). Rather, it constructs the speech from elemental components of a person's recorded voice. This preserves the voice characteristics of the original speaker and hence the system can be easily modified to effect local accents, and other languages. Laureate has been engineered to commercial software standards, is portable between different platforms, and is designed for multichannel operation. Page [3] provides a review of the Laureate architecture and related applications.

In a text-to-speech system (Fig 17), the speech output is derived directly from an unconstrained textual input. The first stage of processing involves expanding the text by a normalisation stage which deals with abbreviations and acronyms, such as 'St. John St.'. This is followed by a stage of syntactic parsing, which aims to resolve ambiguities between words such as 'lives' or 'convict' which have both a verb and a noun form. A semantic analysis may also be carried out which provides a representation of the meaning of text as an aid to word pronunciation. Next, a phonological analysis of the words is made to derive an appropriate word pronunciation, based on a large dictionary and a set of letter-to-sound rules, where the dictionary look-up fails (e.g. for proper nouns). Prosody rules are then applied to provide the required emphasis on duration, intonation and major word stress. Finally the speech is synthesised by looking up the appropriate speech segments, and concatenating them in as seamless a fashion as possible.

Edgington [4] provides a comprehensive review of text processing techniques (the first three blocks in Fig 17). He also provides a review [5] of processing techniques for prosody and waveform generation (the last two blocks in Fig 17).

An example of an application for text-to-speech synthesis is a telephone-based catalogue ordering service, where the system can respond with a full description of the item as well as saying the name and address of where the item is to be sent. Here the quantity of information would be too large to directly record for the average catalogue. A good example of a system where the output information must change rapidly is access to news. Here large amounts of text can be updated regularly and made available over the telephone with no manual intervention. The ability to speak e-mail messages over the telephone is another important application, enabling people to check their messages when away from a computer terminal.

6.5 Speech enhancement

This is the process of enhancing the perceived quality of a speech signal received over a telephone link. This area covers a wide range of applications such as noise and echo control, including the cancellation of echoes on long international circuits to enable simultaneous two-way speech communications. Echoes occur when signals undergo delayed reflections, and can affect both talkers and listeners on a call. The effects of telephony echo on customers are complex, depending on prevailing electrical and acoustic factors, as well as the motivation of both customers [27].

Echo suppressors have been used for many years and conceal echo by detecting when the distant customer is speaking and the near customer is silent.

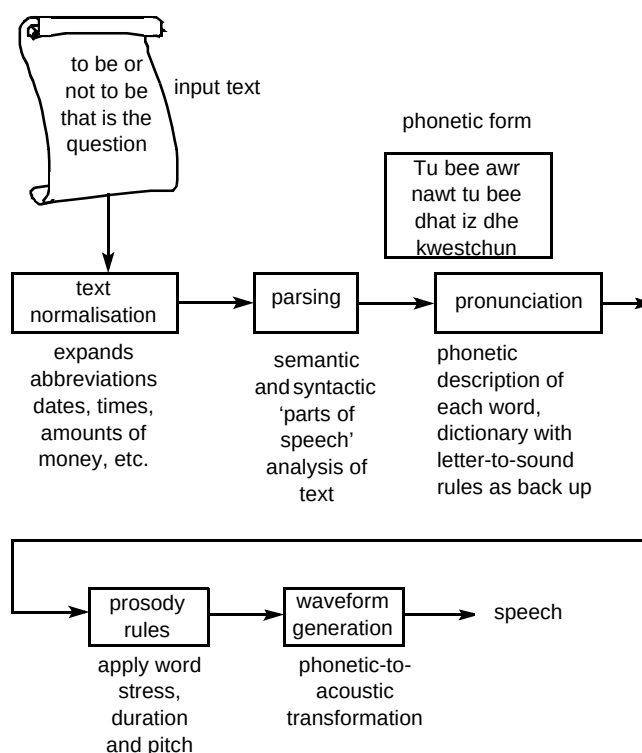


Fig 17 Text-to-speech system.

The alternative of echo cancellation was first proposed by Sondhi of AT&T Bell Labs, and the first single-chip custom-VLSI implementation appeared in 1969. As shown in Fig 18, a replica of the echo signal is synthesised by an adaptive filter which models the echo path and is subtracted from the send signal. In this normal mode of use the echo canceller is only required to model the relatively short duration echoes returned by the national network. In general it is desirable to cancel echoes as close to their source as possible. Cancellers offer somewhat better quality than suppressors on long-delay circuits, because they have greater signal transparency and less intrusive speech clipping effects.

Interest has also grown in applications for adaptive noise suppression where the speech and noise typically have overlapping spectra. In such a system adequate performance is dependent on obtaining a good estimate of the noise

spectrum either from a separate reference microphone, or from noise-only periods in the incoming speech, obtained using a voice-activity detector [2].

Applications include very noisy environments where the signal-to-noise ratio is low, such as telephone kiosks in busy concourses, or in financial dealer rooms. Figure 19 shows an example of a noise suppression system in action, in this case cancelling motorcycle and car noise from a telephone kiosk by the side of a busy road [28].

Signal processing is also employed in speeding up (or slowing down) the playback of recorded messages (without introducing noticeable distortion to the speech), and to eliminate speech-clipping and other nonlinear distortions that are occasionally encountered in public switched telephone networks.

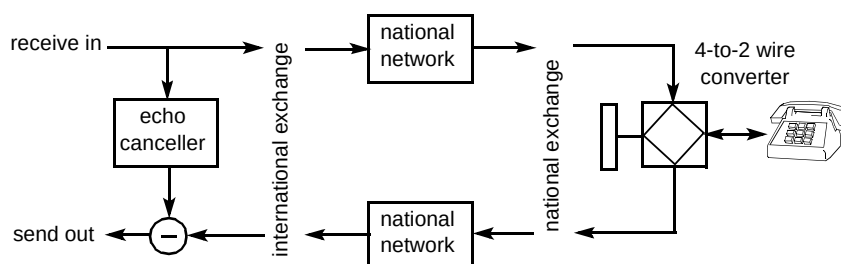


Fig 18 Location of network-based echo cancellers.

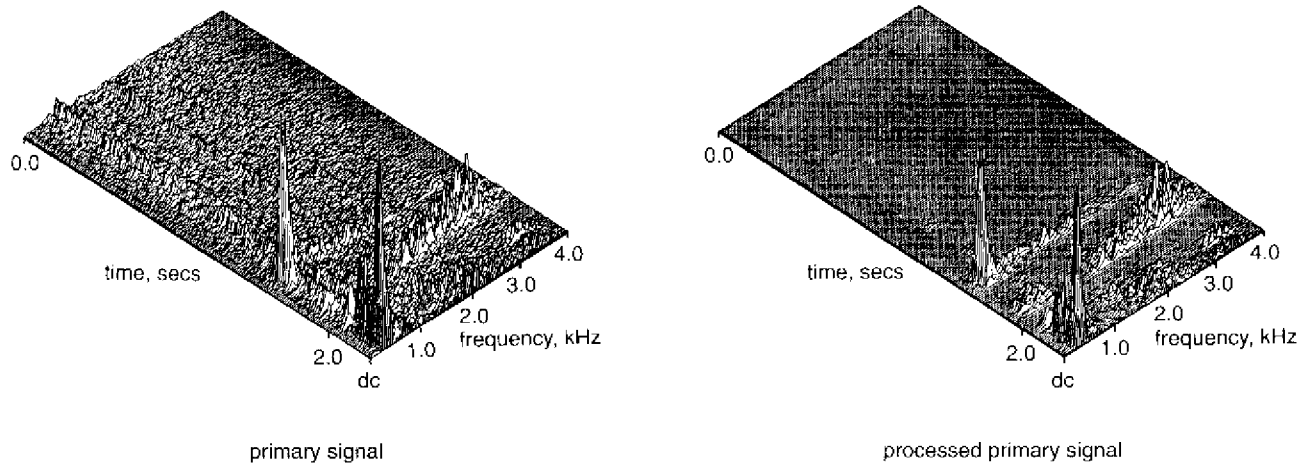


Fig 19 Adaptive noise suppression.

7. Speech technology assessment

Depending on the system being considered, subjective assessment is performed by either a conversation or a listening test [29]. In a listening test, people (who are called subjects) are placed alone in a controlled environment and asked to listen to speech transmitted through a series of randomly presented network connections (which are called conditions). At the end of each condition the subject is asked to vote on their opinion of the speech heard. A 5 point mean-opinion score scale is used (5 = excellent, 4 = good, 3 = fair, 2 = poor, 1 = bad). For example, a speech coder suitable for switched telephone network operation would have a mean-opinion score rating of above 4.0. During a conversation test, two subjects are placed in separate rooms and asked to converse over randomly presented conditions. As an aid to conversation they will be given a simple task to perform. At the end of each condition they each vote on the condition without discussing it with each other. Johnston [9] provides a comprehensive review of subjective testing methods, and describes a novel methodology for comparing text-to-speech systems.

In some cases where the requirements of the final system are not well understood or if a suitably accurate prototype of the recognition sub-system is not available, 'Wizard-of-Oz' experiments are sometimes conducted with a trained human operator in place of the speech recogniser. This allows the focus of the experiment to be maintained on the aspects of dialogue interaction of interest without the complication of errors introduced by the recogniser.

Comparative objective tests are conducted on in-house and commercial speech systems. Figure 20 shows the results of such an assessment on several commercial speech recognisers (averaged results). As can be seen the talker variability is greatest, with the telephone instrument contributing least to recognition errors. Johnston [8]

discusses the testing of speech recognition systems and presents results to show steady progress in reducing error rates as the network has improved.

Recent studies on human perception of speech that has been subjected to nonlinear network distortions have been encouraging, with a high degree of correlation between the objective measures derived and people's subjective opinion of the impairment [14]. There is research interest in extending this modelling approach to joint perception of mixed mode (video and speech) signals for emerging multimedia terminals.

8. Future directions in speech technology

8.1 Interactive voice systems

An essential ingredient in the success of any business is access to information sources and modern telecommunications can give direct access to anywhere in the world within seconds or minutes. With the spread of personal communication products (many of them too small to support a keyboard), speech and language processing, incorporating some degree of understanding, will play a necessary role in providing access to, transforming and delivering information from these vast sources [11, 30].

As interactive speech systems are deployed operationally they will generate new field experience and customers' service perception data. Such 'real-world' data will allow a new R&D focus on technology improvement. Advanced data visualisation techniques will also help researchers to interpret this volume of data. By examining large numbers of interactions, statistical models of dialogues can be generated and used to optimise deployed dialogues that have been developed using traditional techniques. This offers the potential for a unified

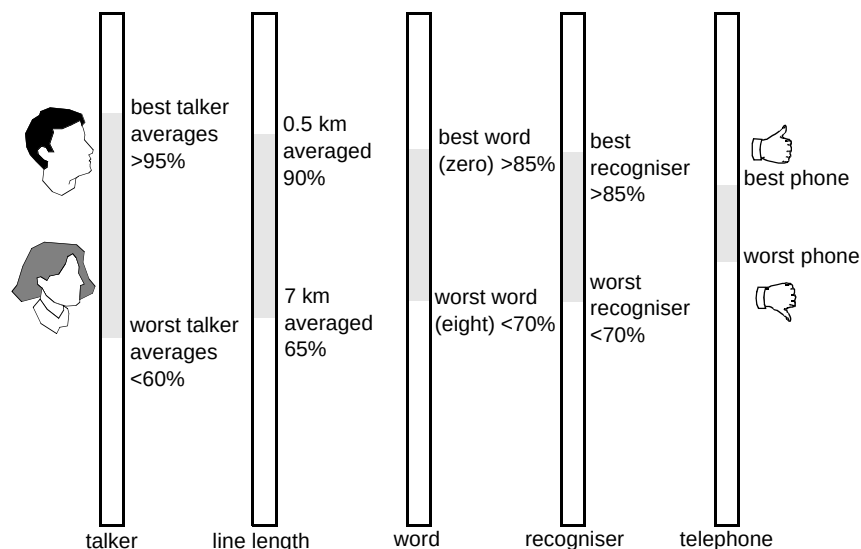


Fig 20 Factors affecting recogniser performance.

understanding from low-level signal processing to higher level semantics.

8.2 Speech coding

As optical-fibre systems provide virtually limitless bandwidth in terrestrial telecommunications networks, the need for speech coding may diminish on circuit-switched point-to-point connections. However, with the growth in corporate 'virtual' networking and wideband packet networks, the added flexibility afforded by variable-rate speech coding will be in demand for competitive managed-bandwidth schemes [1].

Bandwidth will also remain under pressure in mobile and personal communications systems. In such areas, the demand for speech coding will continue to grow strongly over the next decade. This demand will be driven by rising customer expectations on quality, international standardisation, and by technical advances in the on-chip integration of digital and analogue functionality.

There is also considerable interest in using coding techniques to enhance the speech transmitted over networks and for CD-audio compression for future music and multimedia distribution systems.

8.3 Speech synthesis

The current drive to improve the naturalness of text-to-speech systems will continue with the emphasis in areas such as [5]:

- improving the 'natural rhythm' of synthesised speech (phonotactic variation),

- providing more choice in selecting the 'personality' of the speaker (current systems tend to adopt a single speaking style),
- providing more choice of language and accent with the associated tools to provide such flexibility in text-to-speech systems.

As full-motion video requires much bandwidth and/or suffers processing delays, there is growing interest in synthetic model-based image processing coupled with text-to-speech systems. In such a system, a wire-frame model of the person's head is created from photographs and a synthesised voice synchronised to the lips of the model (Fig 21). By typing in text the synthetic persona can be made to 'speak'. Such technology shows promise for new services such as customised database access systems, 'continuous presence' conferencing and to provide a research framework for studying fundamental mixed mode signal interactions.

8.4 Speech recognition

Undoubtedly the current emphasis on accuracy and processing efficiency for large-vocabulary, speaker-independent recognition will continue. To make such systems scalable to national network applications, more work will be needed on improving noise robustness and out-of-vocabulary rejection. These aspects are comprehensively covered in Pawlewski [7]. There is also interest in developing systems that can adapt to the characteristics of the user's voice and the transmission environment. Techniques such as 'barge-in' and 'wordspotting' will also grow in importance in the drive to make speech systems more natural, and hence more acceptable [6, 7].

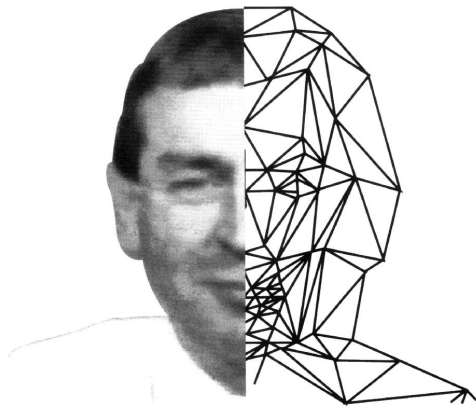


Fig 21 Synthetic persona (talking head).

To enhance the performance of recognisers significantly, more basic knowledge will be needed in areas such as feature extraction, and building in of more knowledge, as it becomes available, of the physiology of the ear and brain as a coupled 'system' (section 4.2). Improvements to modelling of speech can be expected in a drive to reduce the limitations of the existing statistical model-based approaches. New paradigms based on mixed-mode (acoustic, speech, visual and gesture) cues may offer one way forward.

There will also be an increasing interest in multilingual recognition and the associated ability to quickly recognise the language of the speaker. Ultimately, the ability to identify the topic domain as a means of focusing the recognition resources will be developed.

8.5 Person-to-person — the future of telephony

Much of today's speech technology is concerned with man-to-machine interfaces using voice — but what about the future of person-to-person communications? Lewis [15] presents a personal and light-hearted view of the development of the telephone, and its future evolution.

Today, videoconferencing is limited to viewing the participants through a small rectangular window. Images are captured through a single camera and sound by means of a single microphone. In 'hands- and eyes-free' mode the speech often sounds clipped or as if it originated in a bathroom.

There is interest in acoustic echo cancellation, adaptive beam-steering (and talker location) and intelligent loudspeaker technology for 'last metre' processing to remove reverberation and noise in hands-free conferencing systems — to improve the naturalness of such interactions.

Early work in this area at BT Laboratories has shown the potential of this technology to provide improved naturalness and utility for loudspeaking telephones [31].

In the research environment teleconferencing can be made very realistic as with electronic work spaces with eye-to-eye videoconferencing that maintain gaze awareness. When augmented by directional and speaker-tracking audio reproduction the illusion of 'being there' can be almost complete [32].

Speech and language processing has the potential to make such interfaces more natural and easy to use through the exploitation of visual, acoustic and gesture cues. Herein lies the true destiny of speech processing — not just one mode of communications, speech, but all the senses orchestrated to meet the real underlying communications needs of people.

9. Conclusions

Speech communication is, and will continue to be, key to use of the network, currently accounting for well over 90% of revenue. Speech is the most natural for users to communicate from person-to-person and, in the future, where appropriate, from person-to-system. Having the right applications and services is the key to revenue generation. Understanding the technology behind them is vital to designing and operating the network efficiently, to the future of network intelligence and operational support systems, and to reducing costs, while presenting BT as a technologically advanced service provider.

Despite the enormous effort expended on speech technology research over the last few decades, still surprisingly little is known about signals — or rather how humans perceive and interact using them. For example, in areas like speech recognition, much more fundamental knowledge is needed to allow machines to be made that approach the capability of humans. Arguably, there has been too much recent emphasis in the digital signal processing (DSP) community on solving the problems of digital processing and not enough on understanding the subtlety of signals and human perception.

As speech technologies are commercially exploited, different skills and expertise will be required at different stages in their evolution, from research/prototyping in the early stages to downstreaming and support of real-world applications in the later stages. The effective coupling of research to delivery of applications is critical. In particular, the problem of scalability, from laboratory demonstrator to national network service, is fundamental, and one in which BT can be expected to play a key role [13].

Speech processing is a dynamic, exciting and commercially relevant field, overlapping the traditional

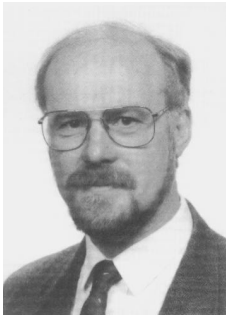
subjects of mathematics, electronics and computer science. Speech technology is well poised to affect dramatically the way we all communicate and interact in the 21st century.

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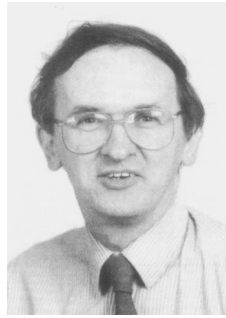
Fred Westall received a BSc(Eng) degree in electrical and electronic engineering from University College of London and the MSc in communication engineering from the University of Manchester, Institute of Science and Technology in 1973 and 1975 respectively. Following a spell in microwave development, he joined BT Laboratories in 1975 to undertake R&D of speechband modems for the public switched telephone network. He has been closely associated with digital signal processing ever since. In 1982 he became head of speech coding, developing novel speech-and-data multiplexers, which incorporated low bit rate and high-quality (7 kHz) speech codecs. In 1987 he became manager of data products development where he was responsible for packet terminals and high-speed modem R&D. Recently he has managed signal processing within the Centre for Human Communications, where he is responsible for downstream speechband applications on DSP speech platforms and for signal processing R&D, in the fields of speech coding, analysis recognition and synthesis. He is a European Engineer, Fellow of the IEE (UK) and Senior Member of the IEEE (USA).



Denis Johnston graduated in electrical engineering at Queen's University, Belfast, in 1971. After a spell in the aerospace industry, he spent two years teaching in Sri Lanka before returning to complete an MSc in electronics and communications. He joined BT in 1974 and has worked in the field of speech technology since then.

Much of his early career was concerned with the assessment of coding methods, channel-sharing techniques and echo control devices for deployment in the telephone network.

In 1983, he became head of a group concerned with the application of speech recognition to telephony products. He is currently a senior advisor in the speech technology unit primarily concerned with evaluation methods.



Alwyn Lewis received the BA degree in electrical sciences from Cambridge University in 1971, and the MSc degree in opto-electronics from Essex University in 1972. After a short spell working for the Plessey Company on formant-tracking coders, he joined BT Laboratories and began work on acoustics, telephonometry and telephone design. In 1978 he helped start a group studying applications of adaptive audio filtering, using hand-wired digital signal processing hardware. He became head of that group in 1984, helping to develop a novel duplex hands-free telephone using a specially

designed VLSI chip-set and a programming DSP microprocessor. In 1987 he moved to the adaptive audio systems group, studying a new DSP technique for intrusive measurements of end-to-end telephony performance. He currently leads the speech coding team of the speech technology unit (part of the Centre for Human Communications), where he has been involved with the development and testing of DSP software for Skyphone™ facsimile and data services. His particular interests include speech coding and enhancement, adaptive audio signal processing, echo cancellation, beam-steering microphones and the social history of electro-technology. He is a Chartered Engineer and a Member of the IEE (UK) and IEEE (USA).